Where a Jacobian-cover decomposition test spends its cost, and the next leads

2026-10-18 · study `research/ic_cover_leads_20261018` · run `run-20261018T1900Z` · build `536aba4ce` (`f4_fp.rs`, `jv_cover.rs` identical to main `3c192c34c`)

Question. After #1717 made `f4_fp` 5× cheaper inside `examples/jv_cover.rs` (#2182), what does one Nagao decomposition test cost now, part by part? Which optimisations come next? Status: **measured** means from this run; **derived** means computed from measured values, with the formula shown; **proposed** means not built.

Answer. At p = 101, C_cov = 1,222,627 field multiplications per test (measured). F4 takes 60 % and the linear stage after it 39 %. Relation collection is 99.99 % of end-to-end S in #2182's dlp records. So per-test savings pass almost one-for-one into S/rho, but only as constant factors.

Measured cost of one test (p = 101, 120 tests)

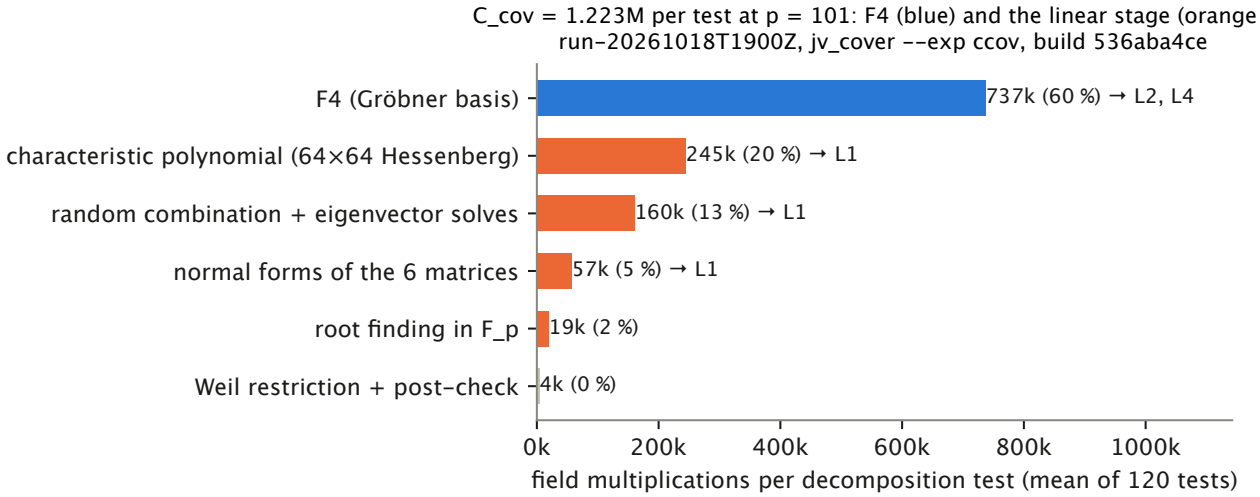


Figure 1: Per-test counted multiplications by part. `jv_cover --exp ccov`, run-20261018T1900Z. The parts add up to the recorded `lin_muls` (481,526) and `c_cov` (1,222,627) means.

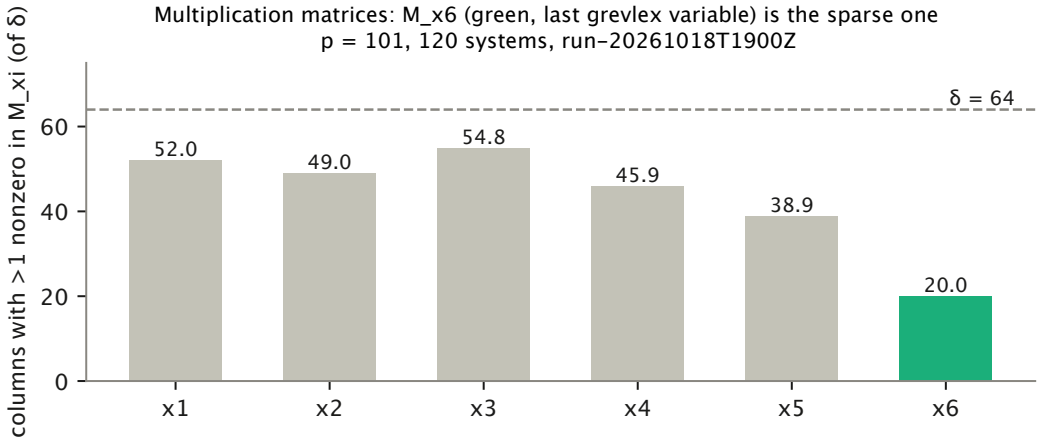


Figure 2: Columns with more than one nonzero entry (mean of 120 systems). `M_x6`, for the last grevlex variable, has 20.0 of δ = 64.

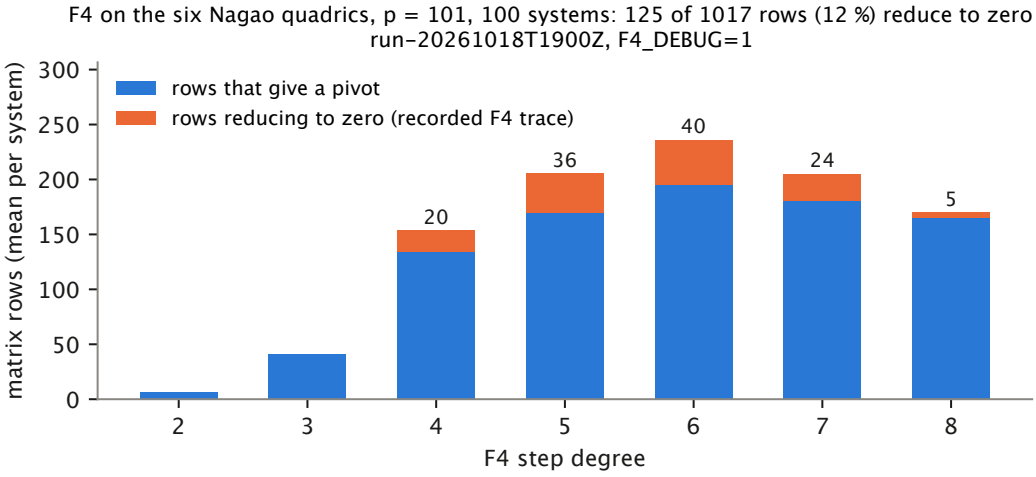


Figure 3: F4 rows per degree over 100 systems: 125 of 1,017 rows (12 %) reduce to zero. These are the Koszul syzygies of a regular sequence.

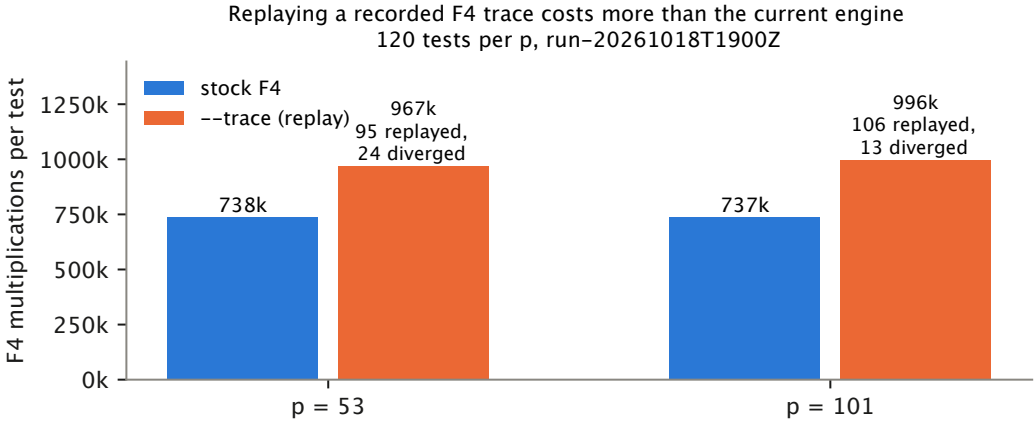


Figure 4: The existing F4 trace replay (`--trace`) costs 31 % (p = 53) and 35 % (p = 101) more F4 multiplications than the stock engine; 24 and 13 of 119 replays diverged.

Leads, ranked

L	lead	change	estimate / status
L1	Sparse-FGLM on `M_x6` (Faugère–Mou)	Build only `M_x6`. Get its minimal polynomial by Wiedemann / Berlekamp–Massey. Explore chain recovery over the 44 columns with at most one nonzero entry; coefficients and connectivity must first be checked. Fall back when x6 does not separate points.	Linear stage 482k → 210k–260k: 2δ·(20δ+44) ≈ 169k + BM 8k + roots 19k + NF 10k–57k + recovery ≈ 3k/root. C_cov ≈ 1.22×–1.29× (derived). Proposed.
L2	F5 / signature criteria	Discard Koszul zero reductions before building them.	125/1,017 recorded rows (12 %) reduced to zero. About 1.08× on C_cov is a row-proportional estimate, not a work bound. Proposed.
L3	Rebuild or retire the F4 trace replay	Record only pivot rows on the current engine, or drop `--trace`.	−23 % to −26 % of F4 multiplications against today's `--trace` (measured regression).
L4	Stop F4 once the needed normal forms are fixed	L1 would narrow normal forms to `M_x6`; the observed 20 multi-entry columns motivate checking the full border set. The degree-7/8 steps hold 37 % of the rows.	Up to 37 % of F4 rows, if a sound early stop exists. Speculative.
L5	Measure S/rho against p	`--exp dlp` at p = 251, 503, 1009, several seeds.	S/rho, mean of the two seeds: 610 (p = 53), 284 (p = 101), slope ≈ p^−1.19 (2 seeds each). Extrapolated, it meets rho between p ≈ 1.7k (p^−2) and p ≈ 12k (p^−1.19); derived and wide. Proposed.

L6	Large-prime variation (Gaudry–Thomé–Thériault–Diem)	Accept decompositions with one or two points outside the factor base and combine them through the large-prime graph.	The only lead here that changes the exponent; not estimated. `jv_cover` has no large-prime path. Proposed.

L1–L4 target constant factors on C_cov. Combining the illustrative L1 and row-proportional L2 estimates gives about 1.4× (1.29 × 1.08); this is unvalidated and is not an upper bound. The distance to rho, 194–850× after #2182, is decided by L5′s slope and L6′s exponent.

Scope and limits

Counted field multiplications only; this run records no wall-time comparisons. The host is a 4 vCPU shared VM (L0). One seed per p for the cost breakdown. The L5 slope rests on two seeds at each of two primes. No change to the engine is made or claimed here.

References

Faugère and Mou, Sparse FGLM, JSC 80 (2017). Faugère, F5, ISSAC 2002. Berthomieu, Eder and Safey El Din, msolve, ISSAC 2021. Gaudry, Thomé, Thériault and Diem, Math. Comp. 76 (2007). Joux and Vitse, Eurocrypt 2012.

Review correction, 2026-10-21: original aggregates reproduced from frozen raw records. The F4 means now divide by 100 systems, rather than the 103 or 98 step records at degrees 6 or 8: 1,017.05 rows, 124.89 zero reductions (12.28%). Corrected derived files are in `reports/review-20261021`. The column counter tests for more than one nonzero entry; it does not certify that every other column is a unit vector. Historical run records remain unchanged. No new performance run was made for this correction.